

```

1 from google.colab import drive
2 drive.mount('/content/drive')
3
4 !pip install fpdf

```

```

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).
Collecting fpdf
  Downloading fpdf-1.7.2.tar.gz (39 kB)
  Preparing metadata (setup.py) ... done
Building wheels for collected packages: fpdf
  Building wheel for fpdf (setup.py) ... done
  Created wheel for fpdf: filename=fpdf-1.7.2-py2.py3-none-any.whl size=40704 sha256=d24ca25756c57a89f7e031541b066d328fff5d2c615
  Stored in directory: /root/.cache/pip/wheels/6e/62/11/dc73d78e40a218ad52e7451f30166e94491be013a7850b5d75
Successfully built fpdf
Installing collected packages: fpdf
Successfully installed fpdf-1.7.2

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```

1 '''
2 getting modified data
3 '''
4
5
6 import copy #we are gonna need this bad boy right here
7 from datasets import Dataset
8 import numpy as np
9 import pandas as pd
10 from sklearn.metrics import accuracy_score
11 import torch
12 from transformers import (
13     BertTokenizer,
14     BertForSequenceClassification,
15     Trainer,
16     TrainingArguments
17 )
18 from torch.utils.data import DataLoader
19 from tqdm.auto import tqdm #add progress bars
20 import os
21
22 def data_prepare():
23     print("prepare data")
24     #name of the multilinli jsonl file
25     file_name = "multilinli_1.0_dev_matched.jsonl"
26     path = "/content/drive/MyDrive/Colab Notebooks/DataforFineTuneBertInstead"
27     #https://pandas.pydata.org/docs/reference/api/pandas.read_json.html I can read this shit line by line with pandas
28
29     entries_full = pd.read_json(path + '/' + file_name, lines = True, encoding = "utf-8")
30     cols = entries_full.columns
31     print(f"name of variables: {cols}")
32     variables_needed = ['gold_label', 'sentence1', 'sentence2']
33
34     #https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.copy.html
35
36     entries = entries_full[variables_needed].copy(deep=True)
37     return entries
38
39
40 def tokenize(entries):
41     print("tokenize")
42     #https://stackoverflow.com/questions/73154119/how-do-i-convert-pandas-dataframe-to-a-huggingface-dataset-object
43     dataset = Dataset.from_pandas(entries)
44     #https://huggingface.co/docs/datasets/v1.8.0/processing.html
45     tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
46
47     #sentence1 = premise
48     #sentence2 = hypothesis
49
50     # This just sentence 1.
51     def tokenize(batch):
52         return tokenizer(
53             batch["sentence1"],
54             truncation=True,
55             padding=False,
56             max_length=128
57         )
58

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59 #https://huggingface.co/docs/datasets/en/about_map_batch
60 dataset = dataset.map(tokenize, batched=True)
61
62 # set pytorch format:
63 #columns = ["input_ids", "attention_mask", "token_type_ids", "label"]
64 #dataset.set_format(type="torch") #remove the columns
65 #print("pytorch format made")
66
67 print("done with tokenization")
68 print(dataset["sentence1"])
69 print(dataset["input_ids"])
70 return dataset, tokenizer
71
72
73 def deswords_tok(tokenizer):
74     words_to_remove = ["on top", "in front", "next to",
75                        "in", "out", "over", "under", "above", "below", "inside",
76                        "outside", "around", "behind", "on", "near", "through",
77                        "throughout", "beneath", "into", "beside", "off", "between",
78                        "next", "upon", "from"]
79
80     words_remove_tok_ids = [tokenizer.encode(word, add_special_tokens = False) for word in words_to_remove]
81     print("Token Ids of words to remove: ", words_remove_tok_ids)
82     return words_remove_tok_ids
83
84
85
86 def find(entries_tok_s1, to_remove):
87     #the code here is halfway between psuedocode and real code it won't run yet.
88
89
90     def contains_sequence(main_sequence, sub_sequence):
91         n = len(sub_sequence)
92         for i in range(len(main_sequence) - n + 1):
93             if main_sequence[i:i+n] == sub_sequence:
94                 return True
95         return False
96
97
98     def filter_logic(data_set):
99         input_ids = data_set['input_ids']
100         for bad_word in to_remove:
101             if contains_sequence(input_ids, bad_word):
102                 return True #keep row
103         return False #no keep
104
105
106 #https://huggingface.co/docs/datasets/en/process #give it some logic and it'll filter rows based on that logic (whateve
107
108 to_be_modified = entries_tok_s1.filter(filter_logic)
109
110 print(len(entries_tok_s1))
111 print(len(to_be_modified))
112
113 return to_be_modified
114
115
116 def find_remove(entries_tok_s1, to_remove, tokenizer):
117
118     #this only filters one entry in the full data set. So I gotta loop through this damn thing for each entry in the dingdar
119     def filter_subseq(to_modify, to_remove):
120         modified = [] #wwhere I save the shit
121         n = len(to_remove) #this is not needed
122         i = 0 #starts off at start at first token in our input string of tokens
123         while i < len(to_modify):
124             matched = False #reset at every removal
125             for token in to_remove: #checks through every banned word in our list of word tokens to remove
126                 m = len(token) #sets the lengt to check to the end of the length of whatever current token string we are looking
127                 if to_modify[i:i+m] == token: #if there is a matching token anywhere that looks like the token we want to remove
128                     matched = True #change the flag
129                     i += m #skip it matching token
130                     break #break the foor loop, but continue the while loop such that we keep checking the rest of the token strir
131             if not matched: #any token that is not on the banned word list enters the modified list, so banned words are exclu
132                 modified.append(to_modify[i])
133                 i += 1 #moves onto next token in string of tokens
134         return modified
135

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136 #adding this filter logic function after spitballing with GeminiAI as it gave me a faster method to converting the fil
137 def filter_logic(batch):
138     new_ids = filter_subseq(batch['input_ids'], to_remove)
139     new_sentence = tokenizer.decode(new_ids, skip_special_tokens = True) #<-- I am not sure if I want to skip these spe
140     return {
141         "input_ids" : new_ids, #I will end up deleting this column. But i'll save it for now just in case
142         "sentence1" : new_sentence #replaces the old sentence1 with the new sentence 1 that should be lacking preposit
143     }
144
145
146     modified_dataset = entries_tok_s1.map(filter_logic)
147
148
149     return modified_dataset
150
151
152 def drop_cols_i_hate(dataset):
153     #https://huggingface.co/docs/datasets/v1.4.0/processing.html#removing-one-or-several-columns-remove-columns
154     #there is a disgusting hair in my keyboard trying to slide under my spacebar
155     dataset.remove_columns(["input_ids", "attention_mask", "token_type_ids"])
156     return dataset
157
158
159 def save_data(modified, file_name):
160
161     #make it pandas
162     modified_df = modified.to_pandas()
163
164     #file path
165     folder_path = "/content/drive/MyDrive/Colab Notebooks/DataforFineTuneBertInstead"
166     os.makedirs(folder_path, exist_ok=True) #makes the file directory if it don't exist
167     file_path = os.path.join(folder_path, file_name + ".jsonl")
168
169     #save it
170     #https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.to_json.html
171     modified_df.to_json(file_path, orient = 'records', lines = True, force_ascii = False)
172     #orient = records makes it so it is in the column, data in columns assorted by rows format. force ascii is false because
173
174     print(f"saved at {file_path}")
175
176     #https://stackoverflow.com/questions/77755675/how-to-convert-huggingface-datasets-to-json-and-vice-versa <-- didn't do
177
178 #silly me I forgot this very important step!
179 def make_copy_of_unmod(data_set):
180     print("haodi")
181     corresponding_unmodified_data = copy.deepcopy(data_set)
182     return corresponding_unmodified_data
183
184
185
186 def main():
187     print("main main")
188     print("main")
189     entries = data_prepare()
190     #https://discuss.huggingface.co/t/from-pandas-dataframe-to-huggingface-dataset/9322
191     #print(entries)
192     #entries = filter_out_hyphen_labels(entries)
193     #print(entries)
194     #entries = gold_label_to_integers(entries) <--- I actually don't want to get rid of hyphens or change labels to integer
195     #print(entries)
196
197
198     #entries_tok_s1 has the same data type as entries
199     #convert sentence 1 to tokens
200     entries_tok_s1, tokenizer = tokenize(entries)
201     print(entries_tok_s1)
202     print(type(entries_tok_s1))
203     print(type(entries_tok_s1["input_ids"]))
204
205     to_remove = deswords_tok(tokenizer)
206     #identify sentence 1s with prepositions; save index number to later put sentence pairs + labels in their own data set
207     #data types of to_be_modified is same as entries_tok_s1 (so it has all the columns, so data shouldn't get fucked up/miss
208     to_be_modified = find(entries_tok_s1, to_remove)
209     #to_be_modified = we have a dataset of only the sentences with select words, but they have not been modified to remove t
210     corresponding_unmodified_set = make_copy_of_unmod(to_be_modified)
211
212     #remove prepositions from that the sentence 1s that were confirmed to have prepositions

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213 #and deconverts sentence 1 back to regular sentence data
214 #and puts the modified sentence 1 back into the running data set,
215 #which means taking input_ids column, making changing it back to words,
216 #shoving it into the sentence 1 column, and then delete the input_ids column to make space for it later,
217 #data structure is still the same
218 modified_s1 = find_remove(to_be_modified, to_remove, tokenizer)
219
220 #the hugging face filter log
221
222 #print(entries['sentence1'])
223 #print(modified_s1["sentence1"])
224
225 print(len(modified_s1)) #the hugging face filter logic prevents this from
226 print(len(entries))
227
228 len_mod = len(modified_s1)
229 len_unmod = len(corresponding_unmodified_set)
230
231 if len_mod != len_unmod:
232     print("lengths of data sets don't match, something is wrong")
233     print(len_mod)
234     print(len_unmod)
235 else:
236     print(len_mod)
237     print(len_unmod)
238     print("the lengths match")
239
240 # and then delete the input_ids column to make space for it later
241 modified_data = drop_cols_i_hate(modified_s1)
242 unmodified_data = drop_cols_i_hate(corresponding_unmodified_set)
243 save_data(modified_data, "modified_multinli_25preps")
244 save_data(unmodified_data, "unmodified_multinli_25preps")
245
246
247 #save it into google drive folder such that it can be called in other Google Colab Cell
248
249
250
251 if __name__ == "__main__":
252     main()

```

```

main main
main
prepare data
name of variables: Index(['annotator_labels', 'genre', 'gold_label', 'pairID', 'promptID',
'sentence1', 'sentence1_binary_parse', 'sentence1_parse', 'sentence2',
'sentence2_binary_parse', 'sentence2_parse'],
dtype='object')
tokenize
Map: 100% 10000/10000 [00:00<00:00, 20180.03 examples/s]
done with tokenization
Column(['The new rights are nice enough', 'This site includes a list of all award winners and a searchable database of Governmen
Column([[101, 1996, 2047, 2916, 2024, 3835, 2438, 102], [101, 2023, 2609, 2950, 1037, 2862, 1997, 2035, 2400, 4791, 1998, 1037,
Dataset({
  features: ['gold_label', 'sentence1', 'sentence2', 'input_ids', 'token_type_ids', 'attention_mask'],
  num_rows: 10000
})
<class 'datasets.arrow_dataset.Dataset'>
<class 'datasets.arrow_dataset.Column'>
Token Ids of words to remove: [[2006, 2327], [1999, 2392], [2279, 2000], [1999], [2041], [2058], [2104], [2682], [2917], [2503]
Filter: 100% 10000/10000 [00:01<00:00, 9430.94 examples/s]
10000
4850
haodi
Map: 100% 4850/4850 [00:01<00:00, 2782.32 examples/s]
4850
10000
4850
4850
the lengths match
saved at /content/drive/MyDrive/Colab Notebooks/DataforFineTuneBertInstead/modified_multinli_25preps.jsonl
saved at /content/drive/MyDrive/Colab Notebooks/DataforFineTuneBertInstead/unmodified_multinli_25preps.jsonl

```

```

1 '''
2 Kyle Urban Capstone
3 as of 1/13/2026 this is my most current issue https://stackoverflow.com/questions/43499199/tensorflow-loss-decreasing-but-a

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4 loss drops from 1.20 to about 0.7 but accuracy never climbs higher than 0.34
5 We are trying one final time here completely fresh
6 1. run this on regular data set (small test)
7 2. run this on regular data set (full test)
8 3. run this on modified data set (full test)
9 4. compare results
10 '''
11 https://huggingface.co/docs/datasets/en/about\_map\_batch
12
13 import gc
14 from datetime import datetime
15 from datasets import Dataset
16 import numpy as np
17 import pandas as pd
18 from sklearn.metrics import accuracy_score
19 import torch
20 from transformers import (
21     BertTokenizer,
22     BertForSequenceClassification,
23     Trainer,
24     TrainingArguments
25 )
26 from torch.utils.data import DataLoader
27 from tqdm.auto import tqdm #add progress bars
28 import os
29 from fpdf import FPDF #gonna make a pdf so I can print this.
30
31
32
33 def compute_metrics(eval_pred):
34     https://datascience.stackexchange.com/questions/31041/what-does-logits-in-machine-learning-mean
35     #logits are like the last scores of the final layer of a model that decide the confidence of whether something is in one
36     logits, labels = eval_pred #unpack the tuple
37     predictions = np.argmax(logits, axis=-1) #this finds the most confident logit (highest/best prediction) of the last layer
38     #the axis = -1 makes the damn thing look at the last layer https://stackoverflow.com/questions/73794766/what-is-the-meaning-of-axis-1
39     return {"accuracy": accuracy_score(labels, predictions)} #should give a column on data table (ngl I did steal this function)
40
41
42 def data_prepare(file_name):
43     print("prepare data")
44     #name of the multinli jsonl file
45     path = "/content/drive/MyDrive/Colab Notebooks/DataforFineTuneBertInstead"
46     https://pandas.pydata.org/docs/reference/api/pandas.read\_json.html I can read this shit line by line with pandas
47
48     entries_full = pd.read_json(path + '/' + file_name, lines = True, encoding = "utf-8")
49     cols = entries_full.columns
50     print(f"name of variables: {cols}")
51     variables_needed = ['gold_label', 'sentence1', 'sentence2']
52
53     https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.copy.html
54
55     entries = entries_full[variables_needed].copy(deep=True)
56     #print(entries)
57     return entries
58
59 def filter_out_hyphen_labels(entries):
60     #note that in the MultiNLI, there is a hyphen gold label as well
61     print("filtering hyphens")
62     https://stackoverflow.com/questions/73907106/pandas-remove-every-entry-with-a-specific-value
63     https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop.html
64     https://pandas.pydata.org/docs/user\_guide/indexing.html
65     print("Unique labels found:", entries['gold_label'].unique())
66
67
68     entries = entries[entries['gold_label'] != '-']
69     return entries
70
71 def gold_label_to_integers(entries):
72     print("map gold_labels to integers")
73     label_map = {'entailment': 2, 'neutral': 1, 'contradiction': 0}
74     entries['label'] = entries['gold_label'].map(label_map)
75
76     https://pandas.pydata.org/docs/reference/api/pandas.Index.dropna.html
77     entries = entries.dropna(subset=['label'])
78     #make sure every label is now an integer
79     entries['label'] = entries['label'].astype(int)
80     return entries
81

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81
82 #so I need to save the original indices
83 #so that with the train unmod set, I can pull them
84 #so I can remove them from them from the eval mod set (which ends up being from the whole set because there is no train_mod
85 def save_og_indices(data_df):
86     print("saving indices")
87     data_df["original_index"] = data_df.index
88
89
90
91 def train_model(entries, num_train_epochs, logging_steps, num_layers_to_freeze, per_device_train_batch_size, per_device_eva
92     print("experiment")
93
94     #adds an extra column that saves the indices
95     save_og_indices(entries)
96
97
98     #https://stackoverflow.com/questions/73154119/how-do-i-convert-pandas-dataframe-to-a-huggingface-dataset-object
99     dataset = Dataset.from_pandas(entries)
100
101
102     dataset = dataset.train_test_split(test_size=0.1)
103     train_dataset = dataset["train"]
104     #I need to steal the train_indices from the train_dataset
105     train_indices = train_dataset['original_index']
106     eval_dataset = dataset["test"]
107     print("train_test_split done")
108
109     #https://huggingface.co/docs/datasets/v1.8.0/processing.html
110     tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
111
112     #sentence1 = premise
113     #sentence2 = hypothesis
114
115     # This concatenates sentence 1 and 2 as a single sequence before getting the CLS embedding.
116     def tokenize(batch):
117         return tokenizer(
118             batch["sentence1"],
119             batch["sentence2"],
120             truncation=True,
121             padding="max_length",
122             max_length=128
123         )
124
125     #https://huggingface.co/docs/datasets/en/about_map_batch
126     train_dataset = train_dataset.map(tokenize, batched=True)
127     eval_dataset = eval_dataset.map(tokenize, batched=True)
128     print("eval and train mapping done")
129
130     # set pytorch format:
131     columns = ["input_ids", "attention_mask", "token_type_ids", "label"]
132     train_dataset.set_format(type="torch") #remove the columns
133     eval_dataset.set_format(type="torch")
134     print("pytorch format made")
135
136
137     # Now for fine-tuning bert:
138     model = BertForSequenceClassification.from_pretrained("bert-base-uncased", num_labels=3)
139     print("model created")
140
141
142     if num_layers_to_freeze != 0:
143         print(f"IMMMMA FREEZIGN THE FIRST: {num_layers_to_freeze} layers")
144         #Freeze bottom N encoder layers (e.g., first 8). This means only the last 4 layers of the model will update.
145         for layer in model.bert.encoder.layer[:num_layers_to_freeze]:
146             for param in layer.parameters():
147                 param.requires_grad = False
148     else:
149         print("no mom it's not cold outside I don't need a jacket")
150
151     print("layers frozen")
152
153     # Training configuration for colab:
154     training_args = TrainingArguments(
155         output_dir="./results",
156         eval_strategy="epoch", #evaluation_strategy got renamed to eval_strategy as per https://github.com/kubeflow/trainer
157         save_strategy="epoch",
158         learning_rate=2e-5,

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159     per_device_train_batch_size=per_device_eval_batch_size, #changing this to see what happens
160     per_device_eval_batch_size=per_device_eval_batch_size,
161     num_train_epochs=num_train_epochs,
162     weight_decay=0.01,
163     load_best_model_at_end= True,
164     metric_for_best_model= "accuracy",
165     save_total_limit = 2,
166     #https://huggingface.co/docs/transformers/v5.0.0/en/main_classes/trainer#transformers.TrainingArguments
167     logging_steps=logging_steps,
168     fp16=fp16, # works well on Colab GPUs allegedly #changing it to false #backt to true for full data set,
169     report_to="none")
170
171 print("training args created")
172
173 #https://huggingface.co/docs/transformers/en/main_classes/trainer
174 #I can add a compute metrics thingy to this
175 # Set up the trainer:
176 trainer = Trainer(
177     model=model,
178     args=training_args,
179     train_dataset=train_dataset,
180     eval_dataset=eval_dataset,
181     processing_class=tokenizer, #had to change because "tokenizer" is now depreciated
182     #https://stackoverflow.com/questions/79546910/typeerror-in-sfttrainer-initialization-unexpected-keyword-argument-toke
183     compute_metrics = compute_metrics
184 )
185
186 print("trainer created")
187 print(train_dataset[0])
188
189 print(entries[['gold_label', 'label']].head(20))
190 print(entries['label'].value_counts())
191
192
193 # Sanity check: ensure that you can overfit a small sample of the data.
194 #https://huggingface.co/docs/transformers/en/main_classes/trainer
195 # I can add metrics such as accuracy score to the trainer (thanks gemini)
196
197 if skip_small == False:
198     trainer.args.num_train_epochs = 50 # Force 50 epochs for the small test
199     small_data_set_size = per_device_train_batch_size + 4
200
201     small_train = train_dataset.shuffle(seed = 0).select(range(small_data_set_size)) #now we are shuffling the deck
202     trainer.train_dataset = small_train
203     trainer.eval_dataset = small_train #I am not sure how this acts as a sanity test
204     trainer.train()
205     results_small = trainer.evaluate()
206     # If this gets close to 100% accuracy, then it's safe to continue. Otherwise there may be bugs.
207     print("small sample set done")
208     print(train_dataset[0]['label'], entries.iloc[0]['label'])
209 else:
210     print("small was skipped")
211     small_data_set_size = 0
212     #I have learned that skipping the overfit of the sample makes performance on the full data set worse. (but does it real
213
214 #1/27/2026 it is possible to overfit the sample set. So we are gonna comment it out.
215
216 # Restore full dataset if overfitting the small sample worked:
217 trainer.args.num_train_epochs = num_train_epochs
218 trainer.train_dataset = train_dataset
219 trainer.eval_dataset = eval_dataset #<--- forgot to add this (will need to rerun in the morning)
220 results_train_full = trainer.train()
221 results_full = trainer.evaluate()
222 print("full set done")
223 print(results_full)
224 print(results_train_full)
225 #print(train_dataset[0]['label'], entries.iloc[0]['label'])
226
227
228
229
230 #return results_full, num_train_epochs, logging_steps, num_layers_to_freeze
231 return trainer, results_full, results_train_full, small_data_set_size, train_indices
232
233 def pandas_to_hug(data_set):
234     #https://stackoverflow.com/questions/73154119/how-do-i-convert-pandas-dataframe-to-a-huggingface-dataset-object
235     dataset_hug = Dataset.from_pandas(data_set)

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```

236     return dataset_nug
237
238 def experiment(data, trainer, train_indices):
239
240     #drop the train pairs used in unmod from the mod to prevent data leakage
241     #data should be a pandas df here
242     data = data.drop(train_indices)
243
244     #converts the modified data set from a pandas df to a hugging face data set
245     data = pandas_to_hug(data)
246
247     #https://huggingface.co/docs/datasets/v1.8.0/processing.html
248     tokenizer = trainer.processing_class
249
250     #sentence1 = premise
251     #sentence2 = hypothesis
252
253
254     #gotta tokenize our modded stuff too... oops forgot to do that
255     # This concatenates sentence 1 and 2 as a single sequence before getting the CLS embedding
256     def tokenize(batch):
257         return tokenizer(
258             batch["sentence1"],
259             batch["sentence2"],
260             truncation=True,
261             padding="max_length",
262             max_length=128
263         )
264
265     data = data.map(tokenize, batched=True)
266
267     #columns = ["input_ids", "attention_mask", "token_type_ids", "label"]
268     data.set_format(type = "torch")
269
270     model = trainer.model
271
272
273     #eval the data set
274     trainer.eval_dataset = data
275     results = trainer.evaluate()
276
277
278     return results
279
280
281 def results_file(trials_do, loss_avg_unmod, accu_avg_unmod, total_results_unmod, loss_avg_mod, accu_avg_mod, total_results_
282                 logging_steps, num_layers_to_freeze, per_device_train_batch_size, per_device_eval_batch_size, skip_small,
283                 loss_avg_train, total_results_train, fp16, small_data_set_size):
284     print("results to file")
285     now = datetime.now()
286     now_str = str(now)
287     folder_path = "/content/drive/MyDrive/Colab Notebooks/DataFromFineTuneBertInstead"
288     os.makedirs(folder_path, exist_ok = True) #makes the ding dang directory if I didn't already make it
289     file_path = os.path.join(folder_path, file_name + now_str + ".txt")
290
291     with open (file_path, 'w') as file:
292         file.write(f"Time: {now_str}\n")
293         file.write(f"Results of modified data eval\n")
294         file.write(f"# of trials: {trials_do}\n")
295         file.write(f"Avg accuracy across trials: {accu_avg_mod}\n")
296         file.write(f"Avg loss across trials: {loss_avg_mod}\n")
297         for i, result in enumerate(total_results_mod):
298             file.write(f"Results from trial #{i + 1}: {result} \n")
299
300         file.write("\n_____ \n")
301
302
303         file.write(f"Results of unmodified data eval\n")
304         file.write(f"# of trials: {trials_do}\n")
305         file.write(f"Avg accuracy across trials: {accu_avg_unmod}\n")
306         file.write(f"Avg loss across trials: {loss_avg_unmod}\n")
307         for i, result in enumerate(total_results_unmod):
308             file.write(f"Results from trial #{i + 1}: {result} \n")
309
310         file.write("\n_____ \n")
311
312         file.write(f"Results of unmodified data training\n")
313         file.write(f"# of trials: {trials_do}\n")

```



```

314 file.write(f"Avg loss across trials: {loss_avg_train}\n")
315 for i, result in enumerate(total_results_train):
316     file.write(f"Results from trial #{i + 1}: {result} \n")
317
318 file.write("\n_____ \n")
319
320
321 file.write(f"Additional Information over training on unmodified data: \n")
322 file.write(f"number of epochs: {num_train_epochs}\n")
323 file.write(f"logging steps: {logging_steps}\n")
324 file.write(f"number of layers frozen: {num_layers_to_freeze}\n")
325 file.write(f"per device train batch size: {per_device_train_batch_size}\n")
326 file.write(f"per device eval batch size: {per_device_eval_batch_size}\n")
327 file.write(f"fp16 status: {fp16}\n")
328
329 if skip_small == False:
330     status_skip = "no"
331 else:
332     status_skip = "yes"
333
334
335 file.write(f"was small sample set (size {small_data_set_size}) training set skipped?: {status_skip}\n")
336
337
338
339
340 # {num_train_epochs, logging_steps, num_layers_to_freeze, per_device_train_batch_size, per_device_eval_batch_size, skip
341
342
343 def results_file_pdf(trials_do, loss_avg_unmod, accu_avg_unmod, total_results_unmod, loss_avg_mod, accu_avg_mod, total_resu
344                     logging_steps, num_layers_to_freeze, per_device_train_batch_size, per_device_eval_batch_size, skip_small,
345                     loss_avg_train, total_results_train, fp16, small_data_set_size):
346     print("gonna do pdf version")
347
348     #this section was stolen from GeminiAI (I did not write this)
349     # 1. Setup the PDF object (Portrait, millimeters, Letter size 8.5x11)
350     pdf = FPDF(orientation='P', unit='mm', format='Letter')
351     pdf.set_auto_page_break(auto=True, margin=15)
352     pdf.add_page()
353     pdf.set_font("Courier", size=10) # Courier mimics code/terminal look nicely
354
355     # Helper function to write lines cleanly
356     def write_line(text, bold=False):
357         if bold:
358             pdf.set_font("Courier", 'B', 10)
359         else:
360             pdf.set_font("Courier", '', 10)
361         # Multi_cell handles text wrapping automatically
362         pdf.multi_cell(0, 5, text)
363         pdf.ln(1) # Add a small gap after lines
364
365     # Helper for divider lines
366     def write_divider():
367         pdf.ln(2)
368         pdf.line(pdf.get_x(), pdf.get_y(), 200, pdf.get_y()) # Draw visual line
369         pdf.ln(4)
370
371
372     def create_results_table(data_list, table_type):
373         if not data_list:
374             pdf.cell(0, 5, "No data available", ln=True)
375             return
376
377         # 1. DEFINE COLUMNS BASED ON TYPE
378         if table_type == "eval":
379             # (Header Name, Dictionary Key, Width)
380             columns = [
381                 ("Trial", None, 15),
382                 ("Loss", "eval_loss", 25),
383                 ("Acc", "eval_accuracy", 25),
384                 ("Runtime", "eval_runtime", 25),
385                 ("Samp/s", "eval_samples_per_second", 25),
386                 ("Steps/s", "eval_steps_per_second", 25),
387                 ("Epoch", "epoch", 20)
388             ]
389         elif table_type == "train":
390             # Different keys for training data

```

```

391     columns = [
392         ("Trial",      None,      15),
393         ("Loss",       "train_loss", 25), # Key inside metrics
394         ("Steps",      "global_step", 20), # Key from object attribute
395         ("Runtime",    "train_runtime", 25),
396         ("Samp/s",     "train_samples_per_second", 25),
397         ("Steps/s",    "train_steps_per_second", 25),
398         ("Epoch",     "epoch",     20)
399     ]
400
401     # 2. Draw Header Row
402     pdf.set_font("Helvetica", 'B', 8) # Smaller font for table
403     for header, key, width in columns:
404         pdf.cell(width, 6, header, border=1, align='C', fill=False)
405     pdf.ln()
406
407     # 3. Draw Data Rows
408     pdf.set_font("Courier", '', 8)
409     for i, raw_result in enumerate(data_list):
410
411         # --- CRITICAL FIX: NORMALIZER LOGIC ---
412         # This converts TrainOutput objects into dictionaries so .get() works
413         row_dict = {}
414         if hasattr(raw_result, 'metrics'): # Checks if it is a TrainOutput object
415             row_dict = raw_result.metrics.copy()
416             row_dict['global_step'] = raw_result.global_step
417             row_dict['train_loss'] = raw_result.training_loss
418             # Map specific keys if needed
419             row_dict['train_runtime'] = row_dict.get('train_runtime')
420             row_dict['train_samples_per_second'] = row_dict.get('train_samples_per_second')
421             row_dict['train_steps_per_second'] = row_dict.get('train_steps_per_second')
422         elif isinstance(raw_result, dict):
423             row_dict = raw_result
424         else:
425             row_dict = {}
426         # -----
427
428         # --- Column 1: Trial Number ---
429         pdf.cell(columns[0][2], 6, str(i + 1), border=1, align='C')
430
431         # --- Columns 2-7: Data from Dictionary ---
432         for header, key, width in columns[1:]:
433             # Get value safely (return '-' if missing)
434             val = row_dict.get(key, '-')
435
436             # Format numbers neatly if they are floats
437             if isinstance(val, float):
438                 val = f"{val:.4f}" # Round to 4 decimal places
439
440             pdf.cell(width, 6, str(val), border=1, align='C')
441
442         pdf.ln() # End of row
443     pdf.ln(5) # Space after table
444
445
446     #I did write this
447     now = datetime.now()
448     now_str = str(now)
449     folder_path = "/content/drive/MyDrive/Colab Notebooks/DataFromFineTuneBertInstead"
450     os.makedirs(folder_path, exist_ok = True) #makes the ding dang directory if I didn't already make it
451     file_path = os.path.join(folder_path, file_name + now_str + ".pdf")
452
453     #this was stolen from Gemini AI
454     # 3. Write Content
455     pdf.set_font("Courier", 'B', 14)
456     pdf.cell(0, 10, "Training Results Report", ln=True, align='C')
457     pdf.set_font("Courier", size=10)
458     pdf.ln(5)
459
460     # Header data
461     write_line(f"Time: {now_str}\n")
462     write_divider()
463
464     # --- Modified Data Eval ---
465     write_line("Results of MODIFIED data eval", bold=True)
466     write_line(f"Trials: {trials_do}")
467     write_line(f"Avg Accuracy: {accu_avg_unmod}")
468     write_line(f"Avg Loss: {loss_avg_mod}")

```

```

468 write_line(f"Avg Loss: {loss_avg_mod} ")
469 pdf.ln(2)
470 create_results_table(total_results_mod, table_type = "eval")
471 write_divider()
472
473 # --- Unmodified Data Eval ---
474 write_line("Results of UNMODIFIED data eval", bold=True)
475 write_line(f"Trials: {trials_do}")
476 write_line(f"Avg Accuracy: {accu_avg_unmod}")
477 write_line(f"Avg Loss: {loss_avg_unmod}")
478 pdf.ln(2)
479 create_results_table(total_results_unmod, table_type = "eval")
480 write_divider()
481
482 # --- Unmodified Data Training ---
483 write_line("Results of UNMODIFIED data training", bold=True)
484 write_line(f"Trials: {trials_do}")
485 write_line(f"Avg Loss: {loss_avg_train}")
486 pdf.ln(2)
487 create_results_table(total_results_train, table_type = "train")
488 write_divider()
489
490 # --- Hyperparameters ---
491 write_line("Configuration & Hyperparameters", bold=True)
492 write_line(f"Epochs: {num_train_epochs}")
493 write_line(f"Logging Steps: {logging_steps}")
494 write_line(f"Frozen Layers: {num_layers_to_freeze}")
495 write_line(f"Train Batch Size: {per_device_train_batch_size}")
496 write_line(f"Eval Batch Size: {per_device_eval_batch_size}")
497 write_line(f"FP16 Status: {fp16}")
498
499 if skip_small == False:
500     status_skip = "no"
501 else:
502     status_skip = "yes"
503
504
505 write_line(f"Small sample set skipped ({small_data_set_size}): {status_skip}")
506 write_divider()
507 write_line("fyi, I did have Gemini AI help me create the function to put my results in this PDF format")
508 write_line("such that it would be more presentable compared to a .txt file")
509
510
511 # 4. Save
512 pdf.output(file_path)
513 print(f"PDF saved to: {file_path}")
514
515
516
517 def trial_maker(unmod_data, mod_data, trials_do, num_train_epochs, logging_steps, num_layers_to_freeze,
518                per_device_train_batch_size, per_device_eval_batch_size, skip_small, fp16, file_name):
519     print("multiple_trials")
520     #since the for loop in this function creates 10 models, I am going to need to clear up memory.
521     total_results_mod = []
522     tally_accu_mod = 0
523     tally_loss_mod = 0
524
525     total_results_unmod = []
526     tally_accu_unmod = 0
527     tally_loss_unmod = 0
528
529     #remember, training is only done on the unmod
530     total_results_train = []
531     tally_accu_train = 0
532     tally_loss_train = 0
533
534
535     for num_trials in range(1, trials_do + 1):
536         #trains model on unmod and does its eval on unmod
537         trainer, results_unmod, results_train, small_data_set_size, train_indices = train_model(unmod_data, num_train_epochs, 1
538                                                  num_layers_to_freeze, per_device_train_batch_size,
539                                                  per_device_eval_batch_size, skip_small, fp16)
540         total_results_unmod.append(results_unmod)
541         tally_loss_unmod += results_unmod['eval_loss']
542         tally_accu_unmod += results_unmod['eval_accuracy']
543
544         #fyi train_indices are the indices of sentence pairs in the unmod that were used to train the model.
545         #they must be used to purge those sentence pairs from the eval of the mod (to prevent data leakage!)

```

```

546
547     total_results_train.append(results_train)
548     tally_loss_train += results_train.metrics['train_loss']
549
550     #evaluates on mod data
551     results_mod = experiment(mod_data, trainer, train_indices)
552     total_results_mod.append(results_mod)
553     tally_loss_mod += results_mod['eval_loss']
554     tally_accu_mod += results_mod['eval_accuracy']
555
556
557     #save partial data incase of a crash
558
559     loss_avg_mod_part = tally_loss_mod/num_trials
560     accu_avg_mod_part= tally_accu_mod/num_trials
561
562     loss_avg_unmod_part = tally_loss_unmod/num_trials
563     accu_avg_unmod_part = tally_accu_unmod/num_trials
564
565     loss_avg_train_part = tally_loss_train/num_trials
566
567     num_trials_s = str(num_trials)
568     trials_do_s = str(trials_do)
569
570     part_name = "completed_" + num_trials_s + "trials out of_" + trials_do_s + "____"
571
572     results_file(num_trials, loss_avg_unmod_part, accu_avg_unmod_part, total_results_unmod, loss_avg_mod_part,
573                 accu_avg_mod_part, total_results_mod, num_train_epochs, logging_steps, num_layers_to_freeze,
574                 per_device_train_batch_size, per_device_eval_batch_size, skip_small, part_name,
575                 loss_avg_train_part, total_results_train, fp16, small_data_set_size)
576
577
578
579     print(f"cleaning memory for trial {num_trials}")
580     del trainer
581     gc.collect()
582     torch.cuda.empty_cache()
583     print(f"Memory cleaned. GPU allocated: {torch.cuda.memory_allocated()}")
584
585
586     loss_avg_mod = tally_loss_mod/trials_do
587     accu_avg_mod = tally_accu_mod/trials_do
588
589     loss_avg_unmod = tally_loss_unmod/trials_do
590     accu_avg_unmod = tally_accu_unmod/trials_do
591
592     loss_avg_train = tally_loss_train/trials_do
593
594     results_file(trials_do, loss_avg_unmod, accu_avg_unmod, total_results_unmod, loss_avg_mod,
595                 accu_avg_mod, total_results_mod, num_train_epochs, logging_steps, num_layers_to_freeze,
596                 per_device_train_batch_size, per_device_eval_batch_size, skip_small, file_name,
597                 loss_avg_train, total_results_train, fp16, small_data_set_size)
598
599     results_file_pdf(trials_do, loss_avg_unmod, accu_avg_unmod, total_results_unmod, loss_avg_mod,
600                     accu_avg_mod, total_results_mod, num_train_epochs, logging_steps, num_layers_to_freeze,
601                     per_device_train_batch_size, per_device_eval_batch_size, skip_small, file_name,
602                     loss_avg_train, total_results_train, fp16, small_data_set_size)
603
604     return 1
605
606 def main():
607     print("main")
608     entries = data_prepare("unmodified_multinli_25preps.jsonl") #these two data sets should be the same size.
609     entries_mod = data_prepare("modified_multinli_25preps.jsonl")
610     #https://discuss.huggingface.co/t/from-pandas-dataframe-to-huggingface-dataset/9322
611     entries = filter_out_hyphen_labels(entries)
612     entries_mod = filter_out_hyphen_labels(entries_mod)
613     entries = gold_label_to_integers(entries)
614     entries_mod = gold_label_to_integers(entries_mod)
615     print(len(entries))
616     print(len(entries_mod))
617     print(entries['sentence1'])
618     print(entries_mod['sentence1'])
619     current_path = "/content/drive/MyDrive/Colab Notebooks/DataforFineTuneBertInstead"
620
621
622     trials_do = 10

```

```
623
624
625 trial_maker(entries, entries_mod, trials_do, 4, 1, 0, 16, 16, True, True, "results")
626 trial_maker(entries, entries_mod, trials_do, 4, 1, 0, 16, 16, False, True, "results")
627
628
629
630 print("done")
631
632
633
634 if __name__ == "__main__":
635     main()
636
637
```



```

main
prepare data
name of variables: Index(['gold_label', 'sentence1', 'sentence2', 'input_ids', 'token_type_ids',
                          'attention_mask'],
                          dtype='object')
prepare data
name of variables: Index(['gold_label', 'sentence1', 'sentence2', 'input_ids', 'token_type_ids',
                          'attention_mask'],
                          dtype='object')
filtering hyphens
Unique labels found: ['contradiction' 'neutral' 'entailment' '-']
filtering hyphens
Unique labels found: ['contradiction' 'neutral' 'entailment' '-']
map gold_labels to integers
map gold_labels to integers
4759
4759
0      well that would be a help i wish they would do...
1      yeah i know and i did that all through college...
2      Calcutta seems to be the only other production...
3      If that investor were willing to pay extra for...
4      The most important directions are simply up an...
...
4845   Second, reducing the rate of HIV transmission ...
4846   Since 1998, LSC has initiated and overseen sig...
4847   Eighty percent of pagers in the United States ...
4848   Finally, the FDA will conduct workshops, issue...
4849   Cirque du Soleil's The latest from the acclaim...
Name: sentence1, Length: 4759, dtype: object
0      well that would be a help i wish they would do...
1      yeah i know and i did that all college and it ...
2      calcutta seems to be the only other production...
3      if that investor were willing to pay extra for...
4      the most important directions are simply up an...
...
4845   second, reducing the rate of hiv transmission ...
4846   since 1998, lsc has initiated and overseen sig...
4847   eighty percent of pagers the united states wer...
4848   finally, the fda will conduct workshops, issue...
4849   cirque du soleil ' s the latest the acclaimed ...
Name: sentence1, Length: 4759, dtype: object
multiple_trials
experiment
saving indices
train_test_split done
Map: 100%                                4283/4283 [00:00<00:00, 6477.79 examples/s]
Map: 100%                                476/476 [00:00<00:00, 6399.43 examples/s]
eval and train mapping done
pytorch format made
Loading weights: 100%                    199/199 [00:00<00:00, 1047.43it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                     | Status |
-----+-----+
cls.predictions.bias                   | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight  | UNEXPECTED |
cls.seq_relationship.weight             | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias               | UNEXPECTED |
cls.predictions.transform.dense.bias    | UNEXPECTED |
classifier.weight                      | MISSING |
classifier.bias                        | MISSING |
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream tas
model created
no mom it's not cold outside I don't need a jacket
layers frozen
training args created
trainer created
{'gold_label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexandr
1999, 23768, 1038, 1012, 1039, 1012, 1010, 10297, 2001, 3007,
1997, 5279, 2076, 1996, 13866, 9890, 2863, 2594, 3690, 1012,
102, 10297, 2001, 2631, 2062, 2084, 2531, 2086, 2044, 1996,
2307, 5233, 1997, 1996, 4833, 2301, 1038, 1012, 1039, 1012,
102, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,

```

[1072/1072 02:22, Epoch 4/4]

Writing model shards: 100%	1/1	[00:00<00:00, 1.36it/s]
Writing model shards: 100%	1/1	[00:00<00:00, 1.42it/s]
Writing model shards: 100%	1/1	[00:00<00:00, 1.40it/s]
Writing model shards: 100%	1/1	[00:00<00:00, 1.39it/s]

```
full set done
{'eval_loss': 0.9292845726013184, 'eval_accuracy': 0.6596638655462185, 'eval_runtime': 0.9845, 'eval_samples_per_second': 483.4}
TrainOutput(global_step=1072, training_loss=0.5821861024987676, metrics={'train_runtime': 142.3067, 'train_samples_per_second'
Map: 100%
476/476 [00:00<00:00, 6293.66 examples/s]
```

```
cleaning memory for trial 1
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train test split done
```

Map: 100% 476/476 [00:00<00:00, 5763.04 examples/s]

Loading weights: 100% 199/199 [00:00<00:00, 827.73it/s, Materializing param=bert.pooler.dense.weight]

Key	Status
cls.predictions.bias	UNEXPECTED
cls.predictions.transform.LayerNorm.bias	UNEXPECTED
cls.predictions.transform.dense.weight	UNEXPECTED


```

cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task model created

no mom it's not cold outside I don't need a jacket

layers frozen

training args created

trainer created

```
{'gold_label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexandr
```

```
1999, 23768, 1038, 1012, 1039, 1012, 1010, 10297, 2001, 3007,
```

```
1997, 5279, 2076, 1996, 13866, 9890, 2863, 2594, 3690, 1012,
```

```
102, 10297, 2001, 2631, 2062, 2084, 2531, 2086, 2044, 1996,
```

```
2307, 5233, 1997, 1996, 4833, 2301, 1038, 1012, 1039, 1012,
```

```
102, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

```
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
```

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0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
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```
gold_label label
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```
0 contradiction 0
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```
1 contradiction 0
```

```
2 neutral 1
```

```
3 contradiction 0
```

```
4 contradiction 0
```

```
5 contradiction 0
```

```
6 neutral 1
```

```
7 contradiction 0
```

```
8 neutral 1
```

```
9 contradiction 0
```

```
10 entailment 2
```

```
11 contradiction 0
```

```
12 entailment 2
```

```
13 entailment 2
```

```
14 neutral 1
```

```
15 neutral 1
```

```
16 contradiction 0
```

```
17 neutral 1
```

```
18 contradiction 0
```

```
19 contradiction 0
```

```
label
```

```
2 1683
```

```
0 1562
```

```
1 1514
```

```
Name: count, dtype: int64
```

```
small was skipped
```

[1072/1072 02:20, Epoch 4/4]

Epoch	Training Loss	Validation Loss	Accuracy
1	0.569730	0.811873	0.647059
2	0.776750	0.800909	0.657563
3	0.399020	0.929199	0.659664
4	0.063135	0.954604	0.647059

Writing model shards: 100% 1/1 [00:00<00:00, 1.42it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.39it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.42it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.43it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',

There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']

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```
Name: count, dtype: int64
small was skipped
```

Epoch	Training Loss	Validation Loss	Accuracy
-------	---------------	-----------------	----------

1	0.569730	0.811873	0.647059
2	0.776750	0.800909	0.657563
3	0.399020	0.929199	0.659664
4	0.063135	0.954604	0.647059

Writing model shards: 100% 1/1 [00:00<00:00, 1.43it/s]

full set done

Map: 100% 476/476 [00:00<00:00, 6274.38 examples/s]

```
cleaning memory for trial 4
```

Memory cleaned. GPU allocated: 18087936

experiment

saving indices

```
train test split done
```

Map: 100% 476/476 [00:00<00:00, 4891.32 examples/s]

```
eval and train mapping done
```

pytorch format made

Loading weights: 100% 199/199 [00:00<00:00, 747.85it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased

Key	Status
cls.predictions.bias	UNEXPECTED
cls.predictions.transform.LayerNorm.bias	UNEXPECTED
cls.predictions.transform.dense.weight	UNEXPECTED
cls.seq_relationship.weight	UNEXPECTED
cls.predictions.transform.LayerNorm.weight	UNEXPECTED
cls.seq_relationship.bias	UNEXPECTED
cls.predictions.transform.dense.bias	UNEXPECTED
classifier.weight	MISSING
classifier.bias	MISSING

[1072/1072 02:18, Epoch 4/4]

Writing model shards: 100%	1/1	[00:00<00:00, 1.45it/s]
Writing model shards: 100%	1/1	[00:00<00:00, 1.41it/s]
Writing model shards: 100%	1/1	[00:00<00:00, 1.41it/s]
Writing model shards: 100%	1/1	[00:00<00:00, 1.43it/s]

7. 1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20. 21. 22. 23. 24. 25. 26. 27. 28. 29. 30. 31. 32. 33. 34. 35. 36. 37. 38. 39. 40. 41. 42. 43. 44. 45. 46. 47. 48. 49. 50. 51. 52. 53. 54. 55. 56. 57. 58. 59. 60. 61. 62. 63. 64. 65. 66. 67. 68. 69. 70. 71. 72. 73. 74. 75. 76. 77. 78. 79. 80. 81. 82. 83. 84. 85. 86. 87. 88. 89. 90. 91. 92. 93. 94. 95. 96. 97. 98. 99. 100. 101. 102. 103. 104. 105. 106. 107. 108. 109. 110. 111. 112. 113. 114. 115. 116. 117. 118. 119. 120. 121. 122. 123. 124. 125. 126. 127. 128. 129. 130. 131. 132. 133. 134. 135. 136. 137. 138. 139. 140. 141. 142. 143. 144. 145. 146. 147. 148. 149. 150. 151. 152. 153. 154. 155. 156. 157. 158. 159. 160. 161. 162. 163. 164. 165. 166. 167. 168. 169. 170. 171. 172. 173. 174. 175. 176. 177. 178. 179. 180. 181. 182. 183. 184. 185. 186. 187. 188. 189. 190. 191. 192. 193. 194. 195. 196. 197. 198. 199. 200. 201. 202. 203. 204. 205. 206. 207. 208. 209. 210. 211. 212. 213. 214. 215. 216. 217. 218. 219. 220. 221. 222. 223. 224. 225. 226. 227. 228. 229. 230. 231. 232. 233. 234. 235. 236. 237. 238. 239. 240. 241. 242. 243. 244. 245. 246. 247. 248. 249. 250. 251. 252. 253. 254. 255. 256. 257. 258. 259. 260. 261. 262. 263. 264. 265. 266. 267. 268. 269. 270. 271. 272. 273. 274. 275. 276. 277. 278. 279. 280. 281. 282. 283. 284. 285. 286. 287. 288. 289. 290. 291. 292. 293. 294. 295. 296. 297. 298. 299. 300. 301. 302. 303. 304. 305. 306. 307. 308. 309. 310. 311. 312. 313. 314. 315. 316. 317. 318. 319. 320. 321. 322. 323. 324. 325. 326. 327. 328. 329. 330. 331. 332. 333. 334. 335. 336. 337. 338. 339. 340. 341. 342. 343. 344. 345. 346. 347. 348. 349. 350. 351. 352. 353. 354. 355. 356. 357. 358. 359. 360. 361. 362. 363. 364. 365. 366. 367. 368. 369. 370. 371. 372. 373. 374. 375. 376. 377. 378. 379. 380. 381. 382. 383. 384. 385. 386. 387. 388. 389. 390. 391. 392. 393. 394. 395. 396. 397. 398. 399. 400. 401. 402. 403. 404. 405. 406. 407. 408. 409. 410. 411. 412. 413. 414. 415. 416. 417. 418. 419. 420. 421. 422. 423. 424. 425. 426. 427. 428. 429. 430. 431. 432. 433. 434. 435. 436. 437. 438. 439. 440. 441. 442. 443. 444. 445. 446. 447. 448. 449. 450. 451. 452. 453. 454. 455. 456. 457. 458. 459. 460. 461. 462. 463. 464. 465. 466. 467. 468. 469. 470. 471. 472. 473. 474. 475. 476. 477. 478. 479. 480. 481. 482. 483. 484. 485. 486. 487. 488. 489. 490. 491. 492. 493. 494. 495. 496. 497. 498. 499. 500. 501. 502. 503. 504. 505. 506. 507. 508. 509. 510. 511. 512. 513. 514. 515. 516. 517. 518. 519. 520. 521. 522. 523. 524. 525. 526. 527. 528. 529. 530. 531. 532. 533. 534. 535. 536. 537. 538. 539. 540. 541. 542. 543. 544. 545. 546. 547. 548. 549. 550. 551. 552. 553. 554. 555. 556. 557. 558. 559. 560. 561. 562. 563. 564. 565. 566. 567. 568. 569. 570. 571. 572. 573. 574. 575. 576. 577. 578. 579. 580. 581. 582. 583. 584. 585. 586. 587. 588. 589. 590. 591. 592. 593. 594. 595. 596. 597. 598. 599. 600. 601. 602. 603. 604. 605. 606. 607. 608. 609. 610. 611. 612. 613. 614. 615. 616. 617. 618. 619. 620. 621. 622. 623. 624. 625. 626. 627. 628. 629. 630. 631. 632. 633. 634. 635. 636. 637. 638. 639. 640. 641. 642. 643. 644. 645. 646. 647. 648. 649. 650. 651. 652. 653. 654. 655. 656. 657. 658. 659. 660. 661. 662. 663. 664. 665. 666. 667. 668. 669. 670. 671. 672. 673. 674. 675. 676. 677. 678. 679. 680. 681. 682. 683. 684. 685. 686. 687. 688. 689. 690. 691. 692. 693. 694. 695. 696. 697. 698. 699. 700. 701. 702. 703. 704. 705. 706. 707. 708. 709. 710. 711. 712. 713. 714. 715. 716. 717. 718. 719. 720. 721. 722. 723. 724. 725. 726. 727. 728. 729. 730. 731. 732. 733. 734. 735. 736. 737. 738. 739. 740. 741. 742. 743. 744. 745. 746. 747. 748. 749. 750. 751. 752. 753. 754. 755. 756. 757. 758. 759. 760. 761. 762. 763. 764. 765. 766. 767. 768. 769. 770. 771. 772. 773. 774. 775. 776. 777. 778. 779. 780. 781. 782. 783. 784. 785. 786. 787. 788. 789. 790. 791. 792. 793. 794. 795. 796. 797. 798. 799. 800. 801. 802. 803. 804. 805. 806. 807. 808. 809. 810. 811. 812. 813. 814. 815. 816. 817. 818. 819. 820. 821. 822. 823. 824. 825. 826. 827. 828. 829. 830. 831. 832. 833. 834. 835. 836. 837. 838. 839. 840.

[1072/1072 02:17, Epoch 4/4]


```
2      neutral      1
3  contradiction    0
4  contradiction    0
5  contradiction    0
6      neutral      1
7  contradiction    0
8      neutral      1
9  contradiction    0
10     entailment    2
11  contradiction    0
12     entailment    2
13     entailment    2
14      neutral      1
15      neutral      1
16  contradiction    0
17      neutral      1
18  contradiction    0
19  contradiction    0
label
2      1683
0      1562
1      1514
Name: count, dtype: int64
small was skipped
[1072/1072 02:17, Epoch 4/4]

Epoch  Training Loss  Validation Loss  Accuracy
1      0.569730      0.811873  0.647059
2      0.776750      0.800909  0.657563
3      0.399020      0.929199  0.659664
4      0.063135      0.954604  0.647059

Writing model shards: 100% 1/1 [00:00<00:00, 1.44it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.41it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.41it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.40it/s]
There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[30/30 00:02]
full set done
{'eval_loss': 0.9292845726013184, 'eval_accuracy': 0.6596638655462185, 'eval_runtime': 0.994, 'eval_samples_per_second': 478.8}
TrainOutput(global_step=1072, training_loss=0.5821861024987676, metrics={'train_runtime': 137.9236, 'train_samples_per_second'
Map: 100% 476/476 [00:00<00:00, 5984.01 examples/s]
results to file
cleaning memory for trial 7
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train_test_split done
Map: 100% 4283/4283 [00:00<00:00, 6342.42 examples/s]
Map: 100% 476/476 [00:00<00:00, 5588.02 examples/s]
eval and train mapping done
pytorch format made
Loading weights: 100% 199/199 [00:00<00:00, 905.21it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream tas
model created
no mom it's not cold outside I don't need a jacket
layers frozen
training args created
trainer created
```


27/62

```
9 contradiction 0
10 entailment 2
11 contradiction 0
12 entailment 2
13 entailment 2
14 neutral 1
15 neutral 1
16 contradiction 0
17 neutral 1
18 contradiction 0
19 contradiction 0
label
2 1683
0 1562
1 1514
Name: count, dtype: int64
small was skipped
[1072/1072 02:17, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.569730 0.811873 0.647059
2 0.776750 0.800909 0.657563
3 0.399020 0.929199 0.659664
4 0.063135 0.954604 0.647059

Writing model shards: 100% 1/1 [00:00<00:00, 1.39it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.42it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.33it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.39it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias'],
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[30/30 00:02]
full set done
{'eval_loss': 0.9292845726013184, 'eval_accuracy': 0.6596638655462185, 'eval_runtime': 1.0059, 'eval_samples_per_second': 473.}
TrainOutput(global_step=1072, training_loss=0.5821861024987676, metrics={'train_runtime': 137.5205, 'train_samples_per_second'
Map: 100% 476/476 [00:00<00:00, 5613.82 examples/s]

results to file
cleaning memory for trial 10
Memory cleaned. GPU allocated: 18087936
results to file
gonna do pdf version
PDF saved to: /content/drive/MyDrive/Colab Notebooks/DataFromFineTuneBertInstead/results2026-02-04 06:23:10.391420.pdf
multiple_trials
experiment
saving indices
train_test_split done

Map: 100% 4283/4283 [00:00<00:00, 6263.32 examples/s]

Map: 100% 476/476 [00:00<00:00, 5685.15 examples/s]

eval and train mapping done
pytorch format made

Loading weights: 100% 199/199 [00:00<00:00, 778.11it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream tas
model created
no mom it's not cold outside I don't need a jacket
layers frozen
training args created
trainer created
{'gold_label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexand
1999, 23768, 1038, 1012, 1039, 1012, 1010, 10297, 2001, 3007,
1997, 5279, 2076, 1996, 13866, 9890, 2863, 2594, 3690, 1012,
102, 10207, 2001, 2631, 2062, 2004, 2521, 2006, 2011, 1006
```


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|----|----------|----------|----------|
| 21 | 0.270020 | 0.218219 | 1.000000 |
| 22 | 0.245819 | 0.198987 | 1.000000 |
| 23 | 0.237549 | 0.180322 | 1.000000 |
| 24 | 0.219543 | 0.163199 | 1.000000 |
| 25 | 0.177643 | 0.148712 | 1.000000 |
| 26 | 0.174652 | 0.136630 | 1.000000 |
| 27 | 0.111786 | 0.126050 | 1.000000 |
| 28 | 0.160782 | 0.117532 | 1.000000 |
| 29 | 0.118454 | 0.107501 | 1.000000 |
| 30 | 0.107681 | 0.099332 | 1.000000 |
| 31 | 0.117752 | 0.093246 | 1.000000 |
| 32 | 0.087769 | 0.086925 | 1.000000 |
| 33 | 0.107330 | 0.080740 | 1.000000 |
| 34 | 0.100128 | 0.074985 | 1.000000 |
| 35 | 0.072807 | 0.069853 | 1.000000 |
| 36 | 0.078941 | 0.065302 | 1.000000 |
| 37 | 0.095558 | 0.061266 | 1.000000 |
| 38 | 0.074265 | 0.057574 | 1.000000 |
| 39 | 0.052322 | 0.054474 | 1.000000 |
| 40 | 0.071396 | 0.051962 | 1.000000 |
| 41 | 0.054939 | 0.049915 | 1.000000 |
| 42 | 0.055389 | 0.048233 | 1.000000 |
| 43 | 0.070824 | 0.046657 | 1.000000 |
| 44 | 0.050957 | 0.045216 | 1.000000 |
| 45 | 0.065956 | 0.044223 | 1.000000 |
| 46 | 0.061058 | 0.043380 | 1.000000 |
| 47 | 0.046860 | 0.042652 | 1.000000 |
| 48 | 0.051643 | 0.042120 | 1.000000 |
| 49 | 0.067345 | 0.041754 | 1.000000 |
| 50 | 0.074287 | 0.041595 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.34it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.45it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.34it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.24it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:02<00:00, 2.11s/it] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.12s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.03it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.24s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |

```
writing model shards: 100% 1/1 [00:00<00:00, 1.06it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.27it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.17it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.37it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.16it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.44it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.12it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.12it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.36it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.42it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.15it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.09it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.32it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.19it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.07s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.30it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.15it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.26it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.24it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.24it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.25it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.12it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.26it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.44it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.31s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.23it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.07s/it]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias'],
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:19, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.511330 0.836416 0.647059
2 1.115040 0.792732 0.680672
3 0.226282 0.936404 0.649160
4 0.271397 1.012029 0.659664

Writing model shards: 100% 1/1 [00:00<00:00, 1.44it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.40it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.37it/s]
```



```
18 contradiction    0
19 contradiction    0
label
2    1683
0    1562
1    1514
Name: count, dtype: int64
```

[100/100 03:12, Epoch 50/50]

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.15s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.05it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.08s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.38it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.04s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.39it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.34it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.09it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.22s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.37it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.02it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.06it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.20it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.39it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.24it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.11it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |

```
Writing model shards: 100% 1/1 [00:00<00:00, 1.12it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.04it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.41it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.05it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.29it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.11it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.03s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.35it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.27it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.43it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.23it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.20s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:19, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.630449 0.807630 0.651261
2 0.795249 0.796047 0.657563
3 0.565233 0.855980 0.651261
4 0.168338 0.891195 0.665966

Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.39it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[30/30 00:02]
full set done
{'eval_loss': 0.8911951184272766, 'eval_accuracy': 0.6659663865546218, 'eval_runtime': 1.0051, 'eval_samples_per_second': 473.
TrainOutput(global_step=1072, training_loss=0.6076614456668273, metrics={'train_runtime': 139.9801, 'train_samples_per_second'
Map: 100% 476/476 [00:00<00:00, 5734.80 examples/s]

results to file
cleaning memory for trial 2
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train_test_split done

Map: 100% 4283/4283 [00:00<00:00, 6434.77 examples/s]

Map: 100% 476/476 [00:00<00:00, 5631.59 examples/s]

eval and train mapping done
pytorch format made

Loading weights: 100% 199/199 [00:00<00:00, 955.03it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream tas
model created
```

```
Name: count, dtype: int64
```

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.20it/s] |

```
Writing model shards: 100% 1/1 [00:00<00:00, 1.27it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.14it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.32it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.19it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.23it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.03it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.31it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.10s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.21it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.29it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.16it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.16it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.08it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.04it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.27it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.14it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.31it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.25it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.24it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.11it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.15it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.13it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.01it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.21it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.19it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.32it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.17it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.42s/it]
Writing model shards: 100% 1/1 [00:01<00:00, 1.07s/it]
Writing model shards: 100% 1/1 [00:01<00:00, 1.00it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.29s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.12it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.21it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.14it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.24it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:18, Epoch 4/4]
```

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 0.630449 | 0.807630 | 0.651261 |
| 2 | 0.795249 | 0.796047 | 0.657563 |
| 3 | 0.565233 | 0.855980 | 0.651261 |
| 4 | 0.168338 | 0.891195 | 0.665966 |

[illegible]

```
10 entailment 2
11 contradiction 0
12 entailment 2
13 entailment 2
14 neutral 1
15 neutral 1
16 contradiction 0
17 neutral 1
18 contradiction 0
19 contradiction 0
label
2 1683
0 1562
1 1514
Name: count, dtype: int64
```



| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.06it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.04s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.25it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.35it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.02it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.41s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.03s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.13it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.08s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.11it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.05it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.11it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.12it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.25it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.01it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.08s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.06it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |

```
Writing model shards: 100% 1/1 [00:00<00:00, 1.35it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.08it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.23it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.12s/it]
Writing model shards: 100% 1/1 [00:01<00:00, 1.47s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.35it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.12it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.36it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.11s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.36it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.09s/it]
Writing model shards: 100% 1/1 [00:01<00:00, 1.15s/it]
Writing model shards: 100% 1/1 [00:01<00:00, 1.96s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.25it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:18, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.630449 0.807630 0.651261
2 0.795249 0.796047 0.657563
3 0.565233 0.855980 0.651261
4 0.168338 0.891195 0.665966

Writing model shards: 100% 1/1 [00:00<00:00, 1.41it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.37it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.27it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.32it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[30/30 00:02]
full set done
{'eval_loss': 0.8911951184272766, 'eval_accuracy': 0.6659663865546218, 'eval_runtime': 1.0264, 'eval_samples_per_second': 463.:
TrainOutput(global_step=1072, training_loss=0.6076614456668273, metrics={'train_runtime': 138.385, 'train_samples_per_second':
Map: 100% 476/476 [00:00<00:00, 5912.44 examples/s]

results to file
cleaning memory for trial 4
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train_test_split done
Map: 100% 4283/4283 [00:00<00:00, 6042.26 examples/s]

Map: 100% 476/476 [00:00<00:00, 5156.54 examples/s]

eval and train mapping done
pytorch format made
Loading weights: 100% 199/199 [00:00<00:00, 883.66it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
```

```
cls.predictions.transform.dense.bias | UNEXPECTED
classifier.weight                    | MISSING
classifier.bias                      | MISSING
```

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

- MISSING : those params were newly initialized because missing from the checkpoint. Consider training on your downstream task with model created

no mom it's not cold outside I don't need a jacket

layers frozen

```
training args created
```

trainer created

```
{'gold label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexandr
```

[illegible]

```
gold_label label
```

contradiction

```
1 contradiction 0
```

2 neutral 1

```
3 contradiction 0
```

4 contradiction 0

```
5 contradiction 0
```

| | | |
|---|---------|---|
| 6 | neutral | 1 |
|---|---------|---|

```
7 contradiction 0
```

| | | |
|---|---------|---|
| 8 | neutral | 1 |
|---|---------|---|

9 contradiction 0

```
10      entailment      2
```

```
11 contradiction 0
```

```
12      entailment      2
```

| | | |
|----|------------|---|
| 13 | entailment | 2 |
|----|------------|---|

| | | |
|----|---------|---|
| 14 | neutral | 1 |
|----|---------|---|

| | | |
|----|---------|---|
| 15 | neutral | 1 |
|----|---------|---|

```
16 contradiction 0
```

| | | |
|----|---------|---|
| 17 | neutral | 1 |
|----|---------|---|

```
18 contradiction 0
```

```
19 contradiction 0
```

label

2 1683

0 1562

1 1514

Name: count, dtype: int64

● [100/100 03:18, Epoch 50/50]

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 12 | 0.731934 | 0.580288 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.12it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.08it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.08s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.11it/s] |

Writing model shards: 100%1/1 [00:01<00:00, 1.40s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.19it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.29it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.21it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.32it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.20it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.03s/it]

Writing model shards: 100%1/1 [00:01<00:00, 1.33s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.27it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.17it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.37it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.25it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.08s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.22it/s]

Writing model shards: 100%1/1 [00:02<00:00, 2.26s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.19it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.08it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.47s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.24it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.09it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.23it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.07s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.25it/s]

Writing model shards: 100%1/1 [00:02<00:00, 2.01s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.20it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.04it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.34s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.07it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.21s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.15it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.00s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.13it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.20s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.29it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.12it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.91s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.20it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.22it/s]

Writing model shards: 100%1/1 [00:01<00:00, 1.17s/it]

Writing model shards: 100%1/1 [00:02<00:00, 2.09s/it]

Writing model shards: 100%1/1 [00:00<00:00, 1.16it/s]

Writing model shards: 100%1/1 [00:00<00:00, 1.35it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']

small sample set done

tensor(0) 0

[1072/1072 02:18, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 0.630449 | 0.807630 | 0.651261 |
| 2 | 0.795249 | 0.796047 | 0.657563 |
| 3 | 0.565233 | 0.855980 | 0.651261 |
| 4 | 0.168338 | 0.891195 | 0.665966 |

Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.30it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.33it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.34it/s]

```
There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma
[30/30 00:02]
```

full set done

```
{'eval_loss': 0.8911951184272766, 'eval_accuracy': 0.6659663865546218, 'eval_runtime': 1.0075, 'eval_samples_per_second': 472.4}
```

```
TrainOutput(global_step=1072, training_loss=0.6076614456668273, metrics={'train_runtime': 138.4313, 'train_samples_per_second'
```

Map: 100% 476/476 [00:00<00:00, 5980.90 examples/s]

```
results to file
```

```
cleaning memory for trial 5
```

```
Memory cleaned. GPU allocated: 18087936
```

experiment

saving indices

train_test_split done

Map: 100% 4283/4283 [00:00<00:00, 6136.97 examples/s]

Map: 100% 476/476 [00:00<00:00, 5202.25 examples/s]

```
eval and train mapping done
```

pytorch format made

Loading weights: 100% 199/199 [00:00<00:00, 1025.04it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased

| Key | Status |
|--|------------|
| cls.predictions.bias | UNEXPECTED |
| cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
| cls.predictions.transform.dense.weight | UNEXPECTED |
| cls.seq_relationship.weight | UNEXPECTED |
| cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
| cls.seq_relationship.bias | UNEXPECTED |
| cls.predictions.transform.dense.bias | UNEXPECTED |
| classifier.weight | MISSING |
| classifier.bias | MISSING |

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream tas model created

no mom it's not cold outside I don't need a jacket

layers frozen

```
training args created
```

```
trainer created
```

```
{'gold_label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexandr
```

[illegible]

gold label label

| | | |
|---|---------------|---|
| 0 | contradiction | 0 |
|---|---------------|---|

| | | |
|---|---------------|---|
| 1 | contradiction | 0 |
|---|---------------|---|

```
2      neutral      1
3  contradiction    0
4  contradiction    0
5  contradiction    0
6      neutral      1
7  contradiction    0
8      neutral      1
9  contradiction    0
10     entailment    2
11  contradiction    0
12     entailment    2
13     entailment    2
14      neutral      1
15      neutral      1
16  contradiction    0
17      neutral      1
18  contradiction    0
19  contradiction    0
label
2    1683
0    1562
1    1514
```

Name: count, dtype: int64



| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.20it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.36it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.09s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.37it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.08s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.11it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.12it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.05s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.88s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.11it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.12s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.18s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.08it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.25it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.29s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.02it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.15it/s] |


```
Writing model shards: 100% 1/1 [00:00<00:00, 1.20it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.26s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.19it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.40it/s]
Writing model shards: 100% 1/1 [00:02<00:00, 2.09s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.20it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.13s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.11it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.24s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.08it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.17s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.17it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.22it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.08it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.01s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.20it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.30it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.15it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.37it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.10it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma
[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:18, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.630449 0.807630 0.651261
2 0.795249 0.796047 0.657563
3 0.565233 0.855980 0.651261
4 0.168338 0.891195 0.665966

Writing model shards: 100% 1/1 [00:00<00:00, 1.37it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.35it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.31it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.34it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma
[30/30 00:02]
full set done
{'eval_loss': 0.8911951184272766, 'eval_accuracy': 0.6659663865546218, 'eval_runtime': 1.0153, 'eval_samples_per_second': 468.1
TrainOutput(global_step=1072, training_loss=0.6076614456668273, metrics={'train_runtime': 138.1731, 'train_samples_per_second'
Map: 100% 476/476 [00:00<00:00, 6165.31 examples/s]
results to file
cleaning memory for trial 6
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train_test_split done
Map: 100% 4283/4283 [00:00<00:00, 6414.91 examples/s]
Map: 100% 476/476 [00:00<00:00, 5414.65 examples/s]
eval and train mapping done
pytorch format made
Loading weights: 100% 199/199 [00:00<00:00, 743.92it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
```


| | | | |
|----|----------|----------|----------|
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.07it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.03it/s] |
| Writing model shards: 100% | 1/1 [00:02<00:00, 2.06s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.34it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.34it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.39it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.51s/it] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.02s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.36it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.05it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.22it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.12it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.13it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.15it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.38it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.34it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.20it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.31it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |

```

[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:18, Epoch 4/4]

```

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 0.630449 | 0.807630 | 0.651261 |
| 2 | 0.795249 | 0.796047 | 0.657563 |
| 3 | 0.565233 | 0.855980 | 0.651261 |
| 4 | 0.168338 | 0.891195 | 0.665966 |

Writing model shards: 100% 1/1 [00:00<00:00, 1.36it/s]

Map: 100% 476/476 [00:00<00:00, 5067.73 examples/s]

Map: 100% 476/476 [00:00<00:00, 5076.84 examples/s]

Loading weights: 100% 199/199 [00:00<00:00, 956.73it/s, Materializing param=bert.pooler.dense.weight]

| Key | Status |
|--|------------|
| cls.predictions.bias | UNEXPECTED |
| cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
| cls.predictions.transform.dense.weight | UNEXPECTED |
| cls.seq_relationship.weight | UNEXPECTED |
| cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
| cls.seq_relationship.bias | UNEXPECTED |
| cls.predictions.transform.dense.bias | UNEXPECTED |
| classifier.weight | MISSING |
| classifier.bias | MISSING |

```
{'gold_label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexandr
```

[illegible]

Name: count, dtype: int64

● [100/100 03:15, Epoch 50/50]

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.200000 | 0.150000 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.04s/it] |
| Writing model shards: 100% | 1/1 [00:02<00:00, 2.03s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.13it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.33it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.14it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.37it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.24it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.40it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.15it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.12s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.24it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.15it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.07it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |

```
Writing model shards: 100% 1/1 [00:00<00:00, 1.07it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.18it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.08it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.10s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.06it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.34it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.01it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.00it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.21it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.00it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.20it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.33it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.16it/s]
Writing model shards: 100% 1/1 [00:01<00:00, 1.14s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.33it/s]
Writing model shards: 100% 1/1 [00:02<00:00, 2.10s/it]
Writing model shards: 100% 1/1 [00:00<00:00, 1.06it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.32it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.01it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.23it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.10it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.26it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.16it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.21it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[2/2 00:33]
small sample set done
tensor(0) 0
[1072/1072 02:18, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.630449 0.807630 0.651261
2 0.795249 0.796047 0.657563
3 0.565233 0.855980 0.651261
4 0.168338 0.891195 0.665966

Writing model shards: 100% 1/1 [00:00<00:00, 1.38it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.32it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.33it/s]
Writing model shards: 100% 1/1 [00:00<00:00, 1.33it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[30/30 00:02]
full set done
{'eval_loss': 0.8911951184272766, 'eval_accuracy': 0.6659663865546218, 'eval_runtime': 1.0049, 'eval_samples_per_second': 473.6}
TrainOutput(global_step=1072, training_loss=0.6076614456668273, metrics={'train_runtime': 138.4586, 'train_samples_per_second'
Map: 100% 476/476 [00:00<00:00, 5126.39 examples/s]

results to file
cleaning memory for trial 8
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train_test_split done
```


4283/4283 | 00:00<00:00, 6077.48 examples/s|

476/476 [00:00<00:00, 5203.85 examples/s]

```
199/199 [00:00<00:00, 845.42it/s, Materializing param=bert.pooler.dense.weight]
```

| Key | Status |
|--|------------|
| cls.predictions.bias | UNEXPECTED |
| cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
| cls.predictions.transform.dense.weight | UNEXPECTED |
| cls.seq_relationship.weight | UNEXPECTED |
| cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
| cls.seq_relationship.bias | UNEXPECTED |
| cls.predictions.transform.dense.bias | UNEXPECTED |
| classifier.weight | MISSING |
| classifier.bias | MISSING |

[illegible]

[100/100 03:23, Epoch 50/50]

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |

| | | | |
|----|----------|----------|----------|
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

| | |
|----------------------------|-----------------------------|
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.19s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.28it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.43s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.12it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.37it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.10s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.01s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.13it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.36s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.02it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.36it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.23it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.26it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.25it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.16it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.09s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.38it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.19it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.29it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.05s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.20it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.27it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.30it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.09it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.21it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.01it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.05s/it] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.17it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.32it/s] |
| Writing model shards: 100% | 1/1 [00:00<00:00, 1.18it/s] |
| Writing model shards: 100% | 1/1 [00:01<00:00, 1.16s/it] |

```
writing model shards: 100% 1/1 [00:00<00:00, 1.11it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.29it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.17it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.14it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.29it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[2/2 00:32]
small sample set done
tensor(0) 0
[1072/1072 02:20, Epoch 4/4]

Epoch Training Loss Validation Loss Accuracy
1 0.630449 0.807630 0.651261
2 0.795249 0.796047 0.657563
3 0.565233 0.855980 0.651261
4 0.168338 0.891195 0.665966

Writing model shards: 100% 1/1 [00:00<00:00, 1.36it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.35it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.28it/s]

Writing model shards: 100% 1/1 [00:00<00:00, 1.30it/s]

There were missing keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.weight', 'bert.embeddings.LayerNorm.bias',
There were unexpected keys in the checkpoint model loaded: ['bert.embeddings.LayerNorm.beta', 'bert.embeddings.LayerNorm.gamma']
[30/30 00:02]
full set done
{'eval_loss': 0.8911951184272766, 'eval_accuracy': 0.6659663865546218, 'eval_runtime': 1.0068, 'eval_samples_per_second': 472.4}
TrainOutput(global_step=1072, training_loss=0.6076614456668273, metrics={'train_runtime': 140.8642, 'train_samples_per_second'

Map: 100% 476/476 [00:00<00:00, 5922.15 examples/s]

results to file
cleaning memory for trial 9
Memory cleaned. GPU allocated: 18087936
experiment
saving indices
train_test_split done

Map: 100% 4283/4283 [00:00<00:00, 6093.41 examples/s]

Map: 100% 476/476 [00:00<00:00, 5293.10 examples/s]

eval and train mapping done
pytorch format made

Loading weights: 100% 199/199 [00:00<00:00, 808.33it/s, Materializing param=bert.pooler.dense.weight]

BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key | Status |
-----+-----+
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream tas
model created
no mom it's not cold outside I don't need a jacket
layers frozen
training args created
trainer created
{'gold_label': 'contradiction', 'sentence1': 'Founded by Alexander the Great on the Mediterranean coast in 322 b.c. , Alexand
1999, 23768, 1038, 1012, 1039, 1012, 1010, 10297, 2001, 3007,
1997, 5279, 2076, 1996, 13866, 9890, 2863, 2594, 3690, 1012,
102, 10297, 2001, 2631, 2062, 2084, 2531, 2086, 2044, 1996,
2307, 5233, 1997, 1996, 4833, 2301, 1038, 1012, 1039, 1012,
102, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0. 0. 0. 0. 0. 0. 0. 0. 0. 0.
```

[100/100 03:13, Epoch 50/50]

| Epoch | Training Loss | Validation Loss | Accuracy |
|-------|---------------|-----------------|----------|
| 1 | 1.022217 | 1.057788 | 0.400000 |
| 2 | 1.079834 | 1.042749 | 0.400000 |
| 3 | 1.045288 | 1.013110 | 0.400000 |
| 4 | 1.150513 | 0.970923 | 0.450000 |
| 5 | 0.830078 | 0.920947 | 0.800000 |
| 6 | 0.917969 | 0.864355 | 0.900000 |
| 7 | 0.841431 | 0.806470 | 0.950000 |
| 8 | 0.887573 | 0.757080 | 1.000000 |
| 9 | 0.826782 | 0.712183 | 1.000000 |
| 10 | 0.828613 | 0.669971 | 1.000000 |
| 11 | 0.714233 | 0.625415 | 1.000000 |
| 12 | 0.731934 | 0.580286 | 1.000000 |
| 13 | 0.797485 | 0.555310 | 1.000000 |
| 14 | 0.436584 | 0.531055 | 1.000000 |
| 15 | 0.576416 | 0.506396 | 1.000000 |
| 16 | 0.517334 | 0.466772 | 1.000000 |
| 17 | 0.504395 | 0.434650 | 1.000000 |
| 18 | 0.406708 | 0.405688 | 1.000000 |
| 19 | 0.420349 | 0.371606 | 1.000000 |
| 20 | 0.305969 | 0.335620 | 1.000000 |
| 21 | 0.461365 | 0.309393 | 1.000000 |
| 22 | 0.398926 | 0.282745 | 1.000000 |
| 23 | 0.316559 | 0.253693 | 1.000000 |
| 24 | 0.379639 | 0.237000 | 1.000000 |

| | | | |
|----|----------|----------|----------|
| 25 | 0.198547 | 0.216727 | 1.000000 |
| 26 | 0.178223 | 0.197311 | 1.000000 |
| 27 | 0.265427 | 0.180493 | 1.000000 |
| 28 | 0.164368 | 0.167242 | 1.000000 |
| 29 | 0.238953 | 0.152127 | 1.000000 |
| 30 | 0.158417 | 0.138116 | 1.000000 |
| 31 | 0.221817 | 0.132710 | 1.000000 |
| 32 | 0.160843 | 0.127109 | 1.000000 |
| 33 | 0.150070 | 0.119128 | 1.000000 |
| 34 | 0.108047 | 0.109653 | 1.000000 |
| 35 | 0.133972 | 0.101006 | 1.000000 |
| 36 | 0.134308 | 0.094769 | 1.000000 |
| 37 | 0.109741 | 0.090161 | 1.000000 |
| 38 | 0.164978 | 0.085773 | 1.000000 |
| 39 | 0.127022 | 0.082587 | 1.000000 |
| 40 | 0.101570 | 0.080118 | 1.000000 |
| 41 | 0.056686 | 0.077777 | 1.000000 |
| 42 | 0.122849 | 0.075964 | 1.000000 |
| 43 | 0.111069 | 0.073579 | 1.000000 |
| 44 | 0.096962 | 0.071645 | 1.000000 |
| 45 | 0.108871 | 0.070724 | 1.000000 |
| 46 | 0.103142 | 0.070216 | 1.000000 |
| 47 | 0.117004 | 0.069919 | 1.000000 |
| 48 | 0.068993 | 0.069426 | 1.000000 |
| 49 | 0.087059 | 0.068988 | 1.000000 |
| 50 | 0.095665 | 0.068755 | 1.000000 |

Writing model shards: 100%

1/1 100:00<00:00 1.26MB/s